Selim Sabag Romero Gonzalez

Postdoctoral Research Associate | Single-Cell Data Core (SCDC)

Texas A&M University | Texas A&M AgriLife Research

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EDUCATION

Ph.D. — Computational Science | The University of Texas at El Paso (UTEP) *2018–2023*

Dissertation: Advances in One-Electron Self-Interaction-Correction Methods for Accurate and Efficient Self-Interaction-Free Density Functional Calculations

M.S. — Computational Science | The University of Texas at El Paso

M.S. — Physics | The University of Texas at El Paso

B.S. — Engineering Physics | Universidad Autónoma de Ciudad Juárez

SUMMARY OF RESEARCH INTERESTS

My research integrates Foundational Quantum Computing with Electrical and Computer Engineering (ECE) applications in complex data analytics and simulation. Drawing on a strong background in theoretical physics and HPC, I specialize in developing and implementing both gate-based quantum algorithms (e.g., Qiskit) and Quantum Annealing (QA) approaches for solving complex, high-dimensional problems. Current work focuses on building novel quantum algorithms (including QA for enhanced feature selection) for classification in single-cell RNA sequencing (scRNA-seq) data, directly addressing challenges in cell differentiation and cancer drug delivery.

EXPERIENCE

Postdoctoral Research Associate | Single-Cell Data Core (SCDC), Texas A&M University *10/2023–Present*

- Develops classical and quantum algorithms to advance data-driven solutions in life sciences and cancer biology.
- Leverages single-cell RNA sequencing and quantum computing for cancer research.
- Conducted in-depth analysis of single-cell data in collaboration with Dr. Chapkin's lab.
- Developing novel quantum algorithms for accurate classification of malignant cells within scRNA-seq data.
- Aggie Research Corps team leader (Fall 2025): 'Machine Learning & Quantum Approaches to Deciphering Gene Signatures'

Graduate Research — Computational Science | The University of Texas at El Paso *2018–2023*

- 5 years of Fortran90 code development for density functional theory (PZSIC, LSIC, FLOSIC).
- Expert in HPC using Fortran90 MPI code; achieved 10–50% speedup with vSOSIC; 80% speedup with parallelized numerical integrations.
- Contributed code patches and implementations to the open-source FLOSIC code.

Teaching & Research Appointments *2015–2023*

- Teaching Assistant — Differential Equations, UTEP (09/2021–10/2023)
- RA appointed by UTEP CoS (Summer 2023, UTEP)
- Research Assistant under Dr. Rajendra Zope, UTEP (06/2020–05/2021)
- Teaching Assistant — MARCS Tutoring Program, UTEP (2018–2019)
- Teaching Assistant — Basic Physics Workshop, UTEP (2016–2018)
- Lecturer, Universidad Autónoma de Ciudad Juárez (06/2015–07/2016)

PUBLICATIONS

- "A Quantum Generative Framework for Modeling Single-Cell Transcriptomes with Gene-Gene and Cell-Cell Interactions", Romero S, Kumar V, Chapkin RS, Cai JJ. arXiv:2510.12776 (2025).

- "Quantum Annealing for Enhanced Feature Selection in Single-Cell RNA Sequencing Data Analysis", Romero S, Gupta S, Gatlin V, Chapkin RS, Cai JJ. Quantum Machine Intelligence (Dec 2025). DOI: 10.1007/S42484-025-00312-1
- "FGF8 induces bone and joint regeneration at digit amputation wounds in neonate mice", Yu L et al., Romero S, Cai JJ. Bone, Vol. 202, Jan 2026. DOI: 10.1016/j.bone.2025.117663
- "Expression of Prooncogenic NR4A-Regulated Genes Inhibited by Dual NR4A1/2 Ligands", Zhang L et al., Romero S. Int. J. Mol. Sci. 2025, 26(8), 3909. DOI: 10.3390/ijms26083909
- "Exploring cell-to-cell variability through differentially variable gene analysis", Gatlin V et al., Romero S. npj Systems Biology and Applications 11, 29 (2025). DOI: 10.1038/s41540-025-00507-z
- "Simplification of the Fermi-Löwdin SIC Method for Efficient DFT Calculations", Romero S, Yamamoto Y, Baruah T, Zope R. ChemPhysChem (Sep 2025). DOI: 10.1002/cphc.202500086
- "Self-consistent implementation of locally scaled self-interaction-correction method", Yamamoto Y et al., Romero S. J. Chem. Phys. 158, 064114 (2023). DOI: 10.1063/5.0130436
- "Spin-state gaps and self-interaction-corrected DFT approximations: Octahedral Fe(II) complexes", Romero S, Baruah T, Zope R. J. Chem. Phys. 158, 054305 (2023). DOI: 10.1063/5.0133999
- "Local self-interaction correction method with a simple scaling factor", Romero S, Yamamoto Y, Baruah T, Zope R. Phys. Chem. Chem. Phys. 2021, 23, 2406. DOI: 10.1039/D0CP06282K
- "A Step in the Direction of Resolving the Paradox of Perdew-Zunger SIC. II.", Bhattarai P et al., Romero S. J. Chem. Phys. 152, 214109 (2020). DOI: 10.1063/5.0010375
- "Improvements in the orbitalwise scaling down of PZ-SIC in many-electron regions", Yamamoto Y, Romero S et al. J. Chem. Phys. 152, 174112 (2020). DOI: 10.1063/5.0004738
- "A GEANT4 Study of a Gamma-ray Collimation Array", Lopez JA, Romero SS et al. J. Nuclear Phys. 7(2), 217-221 (2020). DOI: 10.15415/jnp.2020.72028

PEER REVIEW ACTIVITY

- Quantum Science and Technology — Molecular Docking via Weighted Subgraph Isomorphism on Quantum Annealers (3 rounds, Jun–Sep 2025)
- New Journal of Physics — Thermodynamic Significance of QUBO Encoding on Quantum Annealers (2 rounds, Mar–Apr 2026)
- Machine Learning: Science and Technology — Towards Arbitrary QUBO Optimization (2 rounds, Dec 2024–Mar 2025)

TEACHING RESOURCES & OUTREACH

YouTube Channel: Selim Romero Teaching | Single-Cell RNA Sequencing Tutorial Series*2025–Present*

Founder and instructor of a public YouTube channel providing comprehensive, GUI-based tutorials on single-cell RNA sequencing analysis using scGEAToolbox. Content is designed for beginners and researchers entering the field.

- scGEAToolbox Introduction (25:45)
- Miniconda Installation & Environment Setup (21:56)
- Obtaining Data and Pre-Processing (10:56)
- Differential Expression (DE) Analysis (49:32)
- Differentially Variable (DV) Gene Analysis (28:26)
- Cell Type Gene Markers & Cell Type Annotation (28:26+)

Channel: youtube.com/@SelimRomeroTeaching

PRESENTATIONS

- TAMU Postdoctoral Symposium 2025: QUBO Feature Selection on scRNA-seq — Drug Discovery
- TAMU HPRC Symposium 2025: QUBO Feature Selection on scRNA-seq — Drug Discovery
- CPRIT TAMU Cancer Symposium 2025: QUBO Feature Selection on scRNA-seq — Drug Discovery
- ICIBM 2024: QUBO Feature Selection on scRNA-seq — Cell Differentiation
- TAMU Postdoctoral Symposium 2024: QUBO Feature Selection on scRNA-seq — Cell Differentiation
- CPRIT TAMU Cancer Symposium 2024: Exploring Single-Cell to Cell Variability

- ACS Fall Meeting 2022 (Talk): SCAN Family of Functionals in the FLOSIC Code
- FLOSIC Meeting 2022 (Talk): Selectively Orbital-Scaled SIC for Efficient FLOSIC Calculations
- APS March Meeting 2021 (Talk): Local Scaled SIC Method Using Fermi-Löwdin Orbitals
- UTEP GradEXPO 2021 (Poster): Performance of Local-SIC Using Density Ratio as Scaling Factor
- FLOSIC Meeting 2020 (Poster): Assessment of Local Scaling SIC with a Simpler Scaling Factor
- FLOSIC Meeting 2019 (Poster): Study of Local Scaling SIC with a Simpler Scaling Factor
- APS Fall Texas Section 2017: Geant4 Simulator for Gamma Collimator

CERTIFICATIONS & TRAINING

- Qiskit Global Summer School 2025 — Quantum Excellence (IBM, Aug 2025)
- IOP Trusted Reviewer Status (IOP Publishing, Feb 2025)
- Peer Review Excellence Online Training (IOP Publishing, Dec 2024)
- Qiskit Global Summer School 2024 — Quantum Excellence (IBM, Aug 2024)
- Fundamentals of Accelerated Computing with CUDA C/C++ (NVIDIA, Sep 2024)
- Qiskit Global Summer School 2023: Theory to Implementation (IBM, Jun 2023)
- NERSC Modern Fortran Basics — FUN Training (Jul 2023)
- Codee Training Series: Write Accelerated GPU Code at Expert Level (Apr 2023)
- N-Ways to GPU Programming Bootcamp for NERSC (Apr 2023)
- Intel Developer Tools Optimization Training (Jun 2022)

TECHNICAL SKILLS

- Quantum Computing: Qiskit, Quantum Annealing (D-Wave), QUBO Formulation, VQE, QAOA, QML
- Programming: Fortran90 (expert), Python (expert), Bash/Linux (expert), R, C, CUDA Fortran, Matlab/Octave, Mathematica
- HPC & Parallel Computing: MPI, OpenMP, CUDA, NERSC, TAMU HPRC
- Bioinformatics: Scanpy, Seurat, Signac, scRNA-seq analysis, scikit-learn
- DFT Packages: FLOSIC, NRLMOL, PySCF, PSI4, Gaussian
- Languages: Spanish (Native), English (Professional Proficiency)